

HailGuard Pro

User Guide

Vehicle Hail Damage Intelligence Platform

for Touchless Auto Dent Removal Technology

May 2026

This guide explains how to use HailGuard Pro to explore vehicle hail damage data, identify high-opportunity franchise areas, and support sales & partnership decisions.

1. Overview

HailGuard Pro is a professional-grade intelligence tool designed to help companies offering touchless auto dent removal technology identify high-potential markets, understand historical and recent hail damage patterns, and locate promising business partners or franchise locations.

Key Capabilities

- View 30-year historical hail concentration hotspots with intensity filtering
- Explore specific 2026 severe hail events using a date slider
- See live recent vehicle hail damage reports
- Identify service gap areas (high damage + low repair density)
- View economic impact and business ratings in target regions
- Get AI-powered recommendations on outreach potential per region

Best Use Case: Sales demos, franchise development, and partner outreach in Tornado Alley and the Midwest.

2. Main Interface Overview

The tool consists of three main parts:

1. Interactive Map (Center)

The main visual area. You can pan and zoom freely. All data layers are overlaid here.

2. Sidebar Controls (Left)

Toggle different data layers on/off and adjust sliders for the 30-year and 2026 data.

3. Region Intelligence Panel (Below Map)

This panel automatically updates based on where you are looking on the map. It tells you what data you're seeing, what it means for vehicle damage, and whether the area is a good target for business development.

3. Layer-by-Layer Guide

30-Year Hail Hotspots (Clean Professional View)

This layer shows long-term historical hail damage concentration using clean, professional circle markers instead of blurry heatmaps. Each circle's size and color represent intensity.

- **Slider:** "Minimum Hotspot Intensity" — Drag right to show only the strongest historical zones.
- Higher values = cleaner map focused on the most severe areas.
- **Color Legend:** Red = Very High, Orange = High, Yellow = Moderate-High, etc.

2026 Severe Hail Events (Slider)

This layer lets you view specific severe hail events that occurred in 2026. Use the slider to move through different dates and see exactly where major hailstorms happened.

- Great for showing recent, tangible examples of vehicle damage risk.
- Each event shows hail size and location with a clear popup.

Other Useful Layers

- **Recent Vehicle Hail Reports:** Live data from the last 24 hours (IEM).
- **Service Gaps:** Purple areas = High historical damage but low repair business density (prime franchise targets).
- **Economic Impact:** Orange circles sized by estimated annual hail damage cost.
- **Hail Repair Businesses:** Green/orange/red markers based on Google-style ratings.
- **HRRR Risk Zones:** Areas with >80% probability of severe hail in recent model runs.

4. Using the Region Intelligence Panel (Most Important Feature)

This panel is the heart of the tool for decision-making. It updates automatically as you move the map.

What it shows:

- **Current View Region:** Which major area you are currently looking at.
- **Data Displayed:** Summary of which layers are most relevant in this view.
- **What It Means for Vehicle Hail Damage:** Interpretation of risk level.
- **Outreach Recommendation:** Clear guidance on whether this is a good area for business development.

Recommendation Meanings:

- **Strongly Positive** — High damage + low service density. Excellent franchise or partnership opportunity.
- **Positive (Competitive)** — High damage but more established repair shops. Still good, but more competitive.
- **Positive (High Volume)** — Very high damage volume. Good for high-throughput operations.

Pro Tip: Zoom in on “Strongly Positive” areas and then look at the business markers to find high-rated shops that might be good licensing or franchise partners.

5. Recommended Workflows

For Franchise Development

1. Turn on the 30-Year Hotspots layer.
2. Increase the Intensity slider until only the strongest zones remain.
3. Look at the Region Intelligence panel — focus on “Strongly Positive” areas.
4. Zoom into those areas and turn on the Business layer to find potential partners.

For Sales Demos

1. Start with the 2026 layer and move the slider to show recent dramatic events.
2. Switch to 30-Year view and show the long-term pattern.
3. Highlight Service Gap (purple) areas as untapped opportunity.
4. Use the Region Intelligence panel to give data-backed recommendations.

For Internal Strategy

- Compare multiple regions side-by-side by moving the map and noting the recommendations.
- Use the Economic Impact layer to prioritize highest-ROI markets.

6. Tips & Best Practices

- The map updates live — feel free to pan and zoom while presenting.
- Higher intensity slider values = cleaner, more impactful visuals for executives.
- The 2026 layer is excellent for showing “this is still happening right now.”
- City labels appear above the dots to help with orientation.
- All data is from free public sources (NOAA, NWS, IEM) plus realistic modeling.

This tool was built specifically to support data-driven decisions around touchless dent repair technology deployment, franchising, and strategic partnerships in hail-prone regions.